

* Why Use NC?

- i) Limitation of Synthetic Division $\rightarrow$ root in decimal $\rightarrow$ other than algebraic eq.
- ii) Differential Eqs. which are complex e.g. $\int e^{-x^2} dx$
- iii) System of Linear Eqs. having many unknowns.

$\rightarrow$ An expression of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is called Polynomial of degree n and $a_n \neq 0$

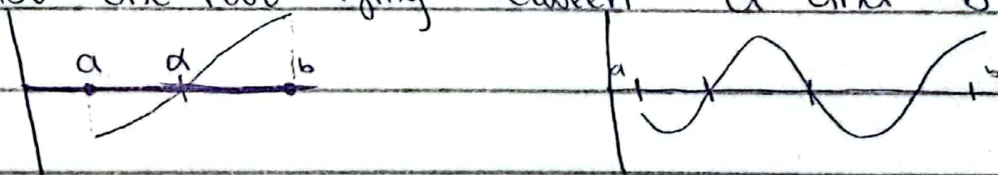
$\rightarrow$ An equation of the form $f(x) = 0$ is called algebraic equation.

$\rightarrow$ An equation that contains exponential, trigonometric or logarithmic function are called transcendental equation.

$\rightarrow$ If α is the root of given equation $f(x) = 0$ then $f(\alpha) = 0$ and α is point on x -axis where graph intersect x -axis

Theorem 8- $\rightarrow$ Limit exist
 $\rightarrow$ no break or discontinuity in graph

$\rightarrow$ If $f(x)$ is continuous on a closed interval $[a, b]$ and $f(a)$ & $f(b)$ are of opposite signs then there is atleast one root lying between a and b .



$\rightarrow$ NC gives approximate answer.

$\rightarrow$ Difference between exact/actual answer and approximate answer is known as error.

$\rightarrow$ Sources of Error :-

- i) Logical Error
- (ii) Truncation Error
- (iii) Precision / Decimal places.

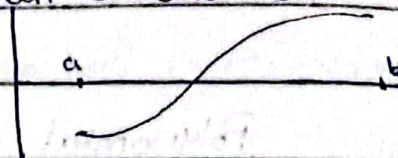
Order of convergence = 1
Bisection Linear

Method 1: BISECTION METHOD

→ Let $f(x)$ be a continuous function on a closed interval $[a, b]$.

→ Let $f(a)$ & $f(b)$ are of opposite signs, then there exist at least one root that lies between a and b .

Here $f(a)$ is -ve &
 $f(b)$ is +ve



* Let 1st Approximation/iteration is $x_1 = \frac{a+b}{2}$

Now if $f(x_1) = 0$, then x_1 is the root. If not then root lies between a and x_1 or x_1 and b

if $f(x_1) < 0$ then root lies between x_1 & b

if $f(x_1) > 0$ then root lies between a & x_1

Let 2nd iteration is $x_2 = \frac{x_1+b}{2}$ (in this case of our equation)

Now if $f(x_2) = 0$ then x_2 is a root. If not then root lies between x_1 and x_2 or x_2 and b .

→ When to stop?

i) When two consecutive iteration have same answer

ii) $f(\text{iteration value}) = 0$. Then iteration value is the root of equation

Bisection Method

→ Always converges
slowly converges
26/1/24

Ex 1: $x^3 + 4x^2 - 10 = 0$ $f(x) = x^3 + 4x^2 - 10$

$f(0) = -10$

$f(1) = -5$

$f(2) = 14$ As $f(1) < 0$ and $f(2) > 0$. So,

root lies between 1 and 2.

The 1st approximation is $x_1 = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$

→ $x_1 = 1.5000$ $f(x_1) = 2.3750$ $f < 0 \vee f > 0$ $f > 0$

As $f(1) < 0$ and $f(x_1) > 0$. So root lies

between 1 and 1.5000

The 2nd approximation is → $x_2 = \frac{1+1.5000}{2} = 1.2500$

$f(x_2) = f(1.2500) = -1.7969$ $f < 0$ $f < 0 \vee f > 0$ $f > 0$

The 3rd iteration is $x_3 = \frac{1.2500+1.5000}{2}$

$x_3 = 1.3750$ $f(x_3) = 0.1621$ $f < 0$ $f > 0$ $f > 0$

The 4th iteration is $x_4 = \frac{1.2500+1.3750}{2} = 1.3125$

$f(x_4) = -0.8484$ $f < 0$ $f < 0$ $f > 0$

$x_5 = \frac{1.3125+1.3750}{2} = 1.3438$

$f(x_5) = -0.3502$ $f < 0$ $f < 0$ $f > 0$

$x_6 = \frac{1.3438+1.3750}{2} = 1.3594$

$f(x_6) = \cancel{0.1003} - 0.0960$ $f < 0$ $f < 0$ $f > 0$

$$x_7 = \frac{1.3594 + 1.3750}{2} = 1.3672$$

$$f(x_7) = 0.0326$$

$f < 0$	$f > 0$	$f > 0$
1.3594	1.3672	1.3750

$$x_8 = \frac{1.3594 + 1.3672}{2} = 1.3633$$

$$f(x_8) = -0.0318$$

$f < 0$	$f < 0$	$f > 0$
1.3594	1.3633	1.3672

$$x_9 = \frac{1.3633 + 1.3672}{2} = 1.3652$$

$$f(x_9) = 0.0012$$

$f < 0$	$f > 0$	$f > 0$
1.3633	1.3652	1.3672

$$x_{10} = 1.3643$$

$$f(1.3643) = -0.0154$$

$$x_{11} = 1.3648$$

$$f(x_{11}) = -0.0071$$

$f < 0$	$f > 0$
1.3633	1.3652

$$x_{12} = 1.3651$$

$$f(x_{12}) = -0.0021$$

1.3643	1.3648	1.3652
--------	--------	--------

$$x_{13} = 1.3652$$

$$f(x_{13}) = -0.0005$$

$$x_{14} = 1.3653$$

$$f(x_{14}) = 0.0012$$

$$x_{15} = 1.3653$$

$$f(x_{15}) = 0.0012$$

So, root of given cubic equation is
 $x = 1.3653$

Ex 2:

$$\cos x = 3x - 1$$

$$f(x) = \cos x - 3x + 1$$

$$f(0) = +2.000$$

$$f(1) = -1.4597$$

0.5	1
-----	---

$$f(x_1) = \frac{-1.4597 + 2.000}{2} = 0.2702$$

$x_1 = \frac{0+1}{2} = 0.5$	$f(x_1) = 0.3776$			
$x_2 = \frac{1+0.5}{2} = 0.75$	$f(x_2) = -0.5183$	$f > 0$	$f > 0$	$f < 0$
$x_3 = \frac{0.5+0.75}{2} = 0.625$	$f(x_3) = -0.0640$	$f > 0$	$f < 0$	$f < 0$
$x_4 = 0.5625$	$f(x_4) = 0.1584$	$f > 0$	$f < 0$	$f < 0$
$x_5 = 0.5938$	$f(x_5) = 0.0474$	$f > 0$	$f < 0$	$f < 0$
$x_6 = 0.6094$	$f(x_6) = -0.0082$	$f > 0$	$f < 0$	$f < 0$
$x_7 = 0.6016$	$f(x_7) = 0.0196$	$f > 0$	$f < 0$	$f < 0$
$x_8 = 0.6055$				

H.W Ex: $3x - e^x = 0$
 $x \log_{10} x - 1.2 = 0$

Order : 1.6

Method of False Position $\left(\begin{array}{l} \text{no possibility of divergence} \\ \text{always converge.} \end{array} \right)$ $1/2/24$

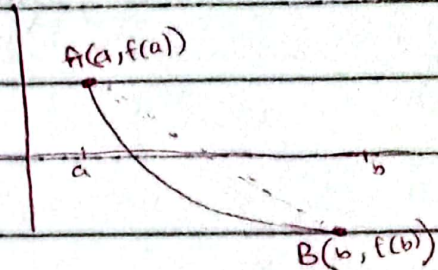
→ Regula Falsi method

→ oldest method of finding real root of $f(x)=0$

→ Similar to Bisection Method.

Consider the equation $f(x)=0$ whose roots are to be determined. Let root of this equation $f(x)=0$ lies between a and b . So this means that $f(a)$ and $f(b)$ are of opposite sign. Then the graph of $y=f(x)$ crosses x -axis at least once in $[a, b]$

The equation of chord joining the points $A(a, f(a))$ & $B(b, f(b))$ is



$$y - y_1 = m(x - x_1)$$

$$y - f(a) = \frac{f(b) - f(a)}{b - a} (x - a) \quad \text{--- (i)}$$

Eg- (i) is a straight line and this 1st line crosses x -axis at x_1 which is called 1st approximation

So $y=0$ $0 - f(a) = \frac{f(b) - f(a)}{b - a} (x_1 - a)$

$$\frac{-f(a)(b-a)}{f(b)-f(a)} = x_1 - a \Rightarrow x_1 = a - \frac{(b-a)f(a)}{f(b)-f(a)}$$

$$x_1 = \frac{a(f(b)-f(a)) - (b-a)f(a)}{f(b)-f(a)}$$

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

If $f(x_1) = 0$, Then x_1 is a root.

If $f(x) \neq 0$, Then root lies between a and x_1 or x_1 & b

If $f(x_1) \leq 0$, So root lies between a and x_1

So 2nd approximation is,

$$x_2 = \frac{af(x_1) - x_1 f(a)}{f(x_1) - f(a)}$$

and so on

Ex Find root of equation $f(x) = x^3 + 4x^2 - 10 = 0$ 4 decimal places

$$f(0) = -10$$

$$f(2) = 14$$

$$f < 0$$

$$f < 0$$

$$f > 0$$

$$f(1) = -5$$

$$a=1, b=2$$

$$1$$

$$1.2632$$

$$2$$

$$x_1 = \frac{1(14) - (-5)(2)}{14 - (-5)} = \frac{4 + 5}{19} = \frac{9}{19} = 1.2632$$

$$f(x_1) = -1.6016 < 0$$

$$f < 0$$

$$f < 0$$

$$f > 0$$

$$1.2632$$

$$1.3388$$

$$2$$

$$x_2 = \frac{1.2632(14) - 2(-1.6016)}{14 - (-1.6016)} = 1.3388$$

$$f(x_2) = -0.4308$$

$$f < 0$$

$$f < 0$$

$$f > 0$$

$$1.3388$$

$$1.3585$$

$$2$$

$$x_3 = \frac{1.3388(14) - 2(-0.4308)}{14 - (-0.4308)} = 1.3585$$

$$f(x_3) = -0.1108$$

$$f < 0$$

$$f < 0$$

$$f > 0$$

$$1.3585$$

$$1.3635$$

$$2$$

$$x_4 = \frac{1.3585(14) - 2(-0.1108)}{14 - (-0.1108)}$$

$$x_4 = 1.3635$$

$$f(x_4) = -0.0285$$

$$N \geq \frac{3 \log_{10} \epsilon}{\log_{10} 2}$$

> sign for slowly convergent methods like Bisection
 = for fast like Regula Falsi

possibility of diversion $\rightarrow$ fastly convergent 2/2/24
 Method of Successive Approximation:-

$\rightarrow$ Also known as iterative method (fixed point method)

Consider the equation $f(x)=0$, whose roots are to be determined. The equation $f(x)=0$ can also be written as

$$x = \phi(x)$$

Then the 1st iteration is

$$x_1 = \phi(x_0) \text{ where } x_0 \text{ is called initial guess}$$

The 2nd iteration is

$$x_2 = \phi(x_1)$$

The 3rd iteration is

$$x_3 = \phi(x_2)$$

$\vdots$

$$x_n = \phi(x_{n-1})$$

If the sequence of iterations $x_1, x_2, x_3, \dots$ converges to some α , then α is called root.

* Here two questions arise.

- i) Does sequence of iterations always converge to α ?
- ii) What is the choice of x_0 ?

Theorem :

If α is the root of $f(x)=0$ and I is the interval in which root exist. Then $\phi(x)$ is defined by eq.

$$x = \phi(x)$$

Then if $|\phi'(x)| < 1, \forall x \in I$

Then sequence of iterations converges to α .

Eg: Find root of equation correct to 7 decimal places.

$$x^3 + x^2 - 1 = 0 \quad f(x) = x^3 + x^2 - 1$$

$$f(0) = -1 < 0 \quad ; \quad f(1) = 1 > 0$$

So root lies between 0 and 1

$$x^3 + x^2 + 1 = 0 \Rightarrow x^2(x+1) = -1$$

$$x^2 = \frac{-1}{1+x} \Rightarrow x = \frac{1}{\sqrt{1+x}}$$

$$\text{As } x = \phi(x)$$

$$\text{So, } \phi(x) = \frac{1}{\sqrt{1+x}}$$

$$\phi'(x) = -\frac{1}{2(1+x)^{3/2}} \quad ; \quad |\phi'(x)| = \frac{1}{2(1+x)^{3/2}}$$

$$\therefore |\phi'(x)| < 1 \quad \forall x \in I \quad \text{So,}$$

$$x_n = \phi(x_{n-1})$$

$$x_n = \frac{1}{\sqrt{1+x_{n-1}}}$$

$$x_1 = \frac{1}{\sqrt{1+x_0}} \quad \text{let } x_0 = 0.65$$

$$x_1 = \frac{1}{\sqrt{1+0.65}} = 0.7498479$$

$$x_2 = \frac{1}{\sqrt{1+x_1}} = 0.7559618$$

$$x_3 = \frac{1}{\sqrt{1+x_2}} = 0.7546446$$

$$x_4 = \frac{1}{\sqrt{1+x_3}} = 0.7549278$$

$$x_5 = \frac{1}{\sqrt{1+x_4}} = 0.7548669$$

$$x_6 = \frac{1}{\sqrt{1+x_5}} = 0.7548800$$

$$x_7 = \frac{1}{\sqrt{1+x_6}} = 0.7548772$$

$$x_8 = \frac{1}{\sqrt{1+x_7}} = 0.7548778$$

$$x_{10} = \frac{1}{\sqrt{1+x_9}} = 0.7548776$$

$$x_{11} = \frac{1}{\sqrt{1+x_{10}}} = 0.7548777$$

$$x_{12} = \frac{1}{\sqrt{1+x_{11}}} = 0.7548777$$

So $\alpha = 0.7548777$ put in eq.

it gives 0.000001

when $q(\alpha) = \alpha$ this is why called fixed point method

Ex :- $2x - \cos x - 3 = 0$ $f(x) = 2x - \cos x - 3$

$$f(0) = -4$$

$$f(1) = -1.54$$

$$f(2) = 1.4 \quad \text{So root between 1 and 2}$$

$$2x - \cos x - 3 = 0$$

$$x = \frac{3 + \cos x}{2} \quad \text{So, } \phi(x) = \frac{3 + \cos x}{2}$$

$$\phi'(x) = \frac{1}{2} (-\sin x); \quad |\phi'(x)| = \frac{\sin x}{2}$$

$$\therefore |\phi'(x)| < 1 \quad \forall x \in I.$$

So, $x_n = \phi(x_{n-1})$

$$x_n = \frac{3 + \cos(x_{n-1})}{2}$$

$$x_1 = \frac{3 + \cos(x_0)}{2} \quad \text{let } x_0 = 1.5$$

$$x_1 = 1.5353686$$

$$x_2 = \frac{3 + \cos(x_1)}{2} = 1.5177102$$

$$x_3 = \frac{3 + \cos(x_2)}{2} = 1.5265306$$

$$x_4 = \frac{3 + \cos(x_3)}{2} = 1.5221256$$

$$x_5 = \frac{3 + \cos(x_4)}{2} = 1.5243257$$

$$\mu_6 = \frac{3 + \cos(\mu_5)}{2} = 1.5232269$$

$$\mu_7 = \frac{3 + \cos(\mu_6)}{2} = 1.5237757$$

$$\mu_8 = 1.5235016$$

$$\mu_9 = 1.5236385$$

$$\mu_{10} = 1.5235702$$

$$\mu_{11} = 1.5236043$$

$$\mu_{12} = 1.5235873$$

$$\mu_{13} = 1.5235958$$

$$\mu_{14} = 1.5235915$$

$$\mu_{15} = 1.5235936$$

$$\mu_{16} = 1.5235926$$

$$\mu_{17} = 1.5235931$$

$$\mu_{18} = 1.5235929$$

$$\mu_{19} = 1.5235930$$

$$\mu_{20} = 1.5235929$$

$$\mu_{21} = 1.5235929$$

$$\alpha = 1.5235929$$

Tangent method

Newton - Raphson

order $\rightarrow 2$ (quadratic)

cannot be used if derivative is 0

approximation can go outside interval 15/2/24

Find the iterative formula for $\sqrt{N}$

$$x = \sqrt{N} \Rightarrow x^2 = N \Rightarrow x^2 - N = 0$$

$$f(x) = x^2 - N ; f'(x) = 2x$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n}$$

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{N}{x_n} \right] \quad \checkmark$$

Eg $\sqrt{12}$

$$x_{n+1} = \frac{1}{2} \left[x_n + \frac{12}{x_n} \right]$$

$$x_1 = \frac{1}{2} \left[x_0 + \frac{12}{x_0} \right]$$

$$\text{let } x_0 = 3.1$$

$$x_1 = 3.4855$$

$$x_2 = \frac{1}{2} \left[x_1 + \frac{12}{x_1} \right] = 3.4642$$

$$x_3 = 3.4641$$

$$x_4 = 3.4641$$

$$x^2 - 12 = 0$$

$$f(0) = -ve$$

$$f(1) = -ve$$

$$f(2) = -ve$$

$$f(3) = -ve \quad \checkmark$$

$$f(4) = +ve$$

Newton rephson method

$$\sin x - 1 - x^3 = 0$$

$$f(x) = \sin x - 1 - x^3$$

$$f(0) = -1$$

$$f(1) = -1.1585$$

$$f(2) = -8.0907$$

$$f(-1) = -2.84147$$

$$f(2) = 6.09070$$

So root is between -1 and -2

$$x_{n+1} = \frac{x_n - f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{\sin(x_n) - 1 - x_n^3}{\cos(x_n) - 3x_n^2}$$

$$= \frac{x_n \cos(x_n) - 3x_n^3 - \sin(x_n) + 1 + x_n^3}{\cos(x_n) - 3x_n^2}$$

$$x_{n+1} = \frac{x_n \cos(x_n) - \sin(x_n) - 2x_n^3 + 1}{\cos(x_n) - 3x_n^2}$$

$$x_1 = \frac{x_0 \cos(x_0) - \sin(x_0) - 2x_0^3 + 1}{\cos(x_0) - 3x_0^2}$$

$$\text{let } x_0 = -1.2$$

$$x_1 = \frac{-1.2 \cos(-1.2) - \sin(-1.2) - 2(-1.2)^3 + 1}{\cos(-1.2) - 3(-1.2)^2}$$

$$x_1 = -1.25156$$

$$x_2 = \frac{x_1 \cos(x_1) - \sin(x_1) - 2(x_1)^3 + 1}{\cos(x_1) - 3(x_1)^2}$$

$$x_2 = -1.24906$$

$$x_3 = \frac{x_2 \cos(x_2) - \sin(x_2) - 2(x_2)^3 + 1}{\cos(x_2) - 3(x_2)^2}$$

$$x_3 = -1.24905$$

$$x_4 = \frac{x_3 \cos(x_3) - \sin(x_3) - 2(x_3)^3 + 1}{\cos(x_3) - 3(x_3)^2}$$

$$x_4 = -1.24905$$

$$\text{So root } x = -1.24905$$

* Find the iterative formula for $\sqrt[p]{N}$ or $(N)^{1/p}$

$$x = \sqrt[p]{N} \Rightarrow x^p = N \Rightarrow x^p - N = 0$$

$$f(x) = x^p - N \quad \therefore \quad f'(x) = p x^{p-1}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^p - N}{p x_n^{p-1}}$$

$$x_{n+1} = \frac{p x_n^p - x_n^p + N}{p x_n^{p-1}}$$

$$= \frac{x_n^p (p-1) + N}{p x_n^{p-1}} = \frac{x_n^p (p-1)}{p x_n^{p-1}} + \frac{N}{p x_n^{p-1}}$$

$$= \frac{p-1}{p} x_n + \frac{N}{p} x_n^{1-p}$$

$$= \left(1 - \frac{1}{p}\right) x_n + \frac{N}{p} x_n^{1-p}$$

ex. $(25)^{1/3}$

$$N = 25, \quad p = 3$$

$$x_{n+1} = \left(1 - \frac{1}{3}\right) x_n + \frac{25}{3} x_n^{1-3}$$

$$= \frac{2}{3} x_n + \frac{25}{3} x_n^{-2} = \frac{2}{3} x_n + \frac{25}{3} \frac{1}{x_n^2}$$

$$x_{n+1} = \frac{2x_n^3 + 25}{3x_n^2}$$

$$x_1 = \frac{2x_0^3 + 25}{3x_0^2}$$

$$\text{let } x_0 = 2.9$$

$$x_1 = 2.9242$$

$$x_2 = 2.9240$$

$f(x) = x^3 - 25$
 $f(0) = -ve$
 $f(1) = -ve$
 $f(2) = -ve$
 $f(3) = -ve$
 $f(4) = +ve$

$x \approx 2.9240$

$x + \frac{1}{p}x$
 $p x - 1 x$
 $\frac{1}{x} - p$ ✓ $\rightarrow$ bcz derivative removes variable

Find iterative formula for $\frac{1}{p}$

$$x = \frac{1}{p} \Rightarrow \frac{1}{x} - p = 0$$

$$f(x) = \frac{1}{x} - p \quad ; \quad f'(x) = -\frac{1}{x^2}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{\frac{1}{x_n} - p}{-\frac{1}{x_n^2}}$$

$$x_{n+1} = x_n (2 - p x_n) \quad \checkmark$$

ex. e^{-1}

$$x_{n+1} = x_n (2 - e x_n)$$

$$x_1 = x_0 (2 - e x_0)$$

$$\text{let } x_0 = 0.3$$

$$x_2 = 0.3675$$

$$x_3 = 0.3679$$

$$x_4 = 0.3679$$

$$\therefore \text{So } \alpha = 0.3679$$

16/2/24

SECCANT METHOD $\rightarrow$ fastly convergent $\rightarrow$ can diverge also.
 $\rightarrow$ order: 1.618...

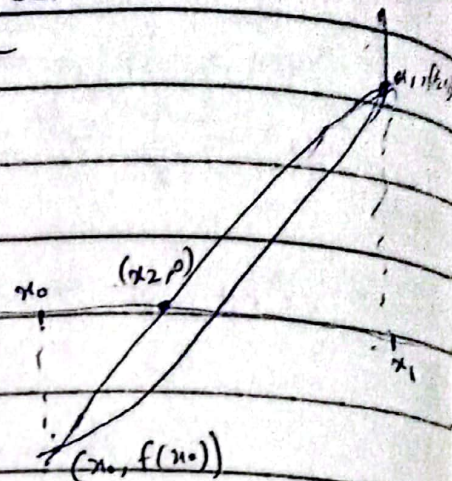
$\rightarrow$ Similar to regula falsi Method.

$y - y_0 = m(x - x_0)$ $\rightarrow$ can take x_0 or x_1

$y - f(x_0) = m(x - x_0)$

$y - f(x_0) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$

$0 - f(x_0) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x_2 - x_0)$



~~$f(x_0) = f(x_1) - f(x_0)$~~ $x_2 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$

Similarly, $x_3 = x_1 - \frac{f(x_1)(x_2 - x_1)}{f(x_2) - f(x_1)}$

Generally, $x_{n+1} = x_{n-1} - \frac{f(x_{n-1})(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$

or

$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$

$= \frac{x_{n-1}f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$

ex. $xe^x = \cos x$

$$x_{n+1} = x_n - \frac{f(x_n) [x_n - x_{n-1}]}{f(x_n) - f(x_{n-1})}$$

$$f(x) = xe^x - \cos x$$

$$f(0) = -1$$

$$; f(1) = 2.178$$

let suppose $x_0 = 0.2$ and $x_1 = 0.3$

$$x_2 = x_1 - \frac{f(x_1) [x_1 - x_0]}{f(x_1) - f(x_0)}$$

$$= 1 - \frac{(Ae^A - \cos A) [A - B]}{(Ae^A - \cos A) - (Be^B - \cos B)}$$

$$x_2 = 0.5968$$

$$x_3 = 0.5024$$

$$x_4 = 0.5168$$

$$x_5 = 0.5178$$

$$x_6 = 0.5178$$

• If at any iteration $f(x_n) = f(x_{n-1})$
then this method fails and shows
that it does not necessarily converge
This is a drawback of secant method over
regula falsi method.

22/2/24

→ Exact Method → only applicable if solution is unique

Method of Factorization

OR

LU-Decomposition Method

Consider a system of Linear Eq.

$$\left. \begin{array}{l} 3x + y - z = 3 \\ 2x - 8y + z = -5 \\ x - 2y + 9z = 8 \end{array} \right\} \text{--- (1) This system of Linear Eq. can be written as}$$

$$AX = b \text{ --- (2)}$$

Let $LU = A$

where $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$, $U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$

So $\Rightarrow$ (2)

→ Unit lower triangular

→ upper triangular

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} U_{11} & U_{12} & U_{13} \\ l_{21}U_{11} & l_{21}U_{12} + U_{22} & l_{21}U_{13} + U_{23} \\ l_{31}U_{11} & l_{31}U_{12} + l_{32}U_{22} & l_{31}U_{13} + l_{32}U_{23} + U_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

By Comparing

$$U_{11} = 3$$

$$U_{22} = 1$$

$$U_{33} = -1$$

$$l_{21}U_{11} = 2$$

$$l_{21}U_{12} + U_{22} = -8$$

$$l_{21}U_{13} + U_{23} = 1$$

$$l_{31}U_{11} = 1$$

$$l_{31}U_{12} + l_{32}U_{22} = -2$$

$$l_{31}U_{13} + l_{32}U_{23} + U_{33} = 9$$

→ if any value of diagonal entry of U is 0 then free variable occurs. Determinant is 0 and method not applicable

$$\Rightarrow l_{21} = 2/3$$

$$\Rightarrow l_{31} = 1/3$$

$$\Rightarrow U_{22} = -8 - 2/3 = -\frac{26}{3}$$

$$\Rightarrow U_{23} = 1 + 2/3 = 5/3$$

$$\Rightarrow l_{32} = \frac{-2 - 1/3}{-26/3} = \frac{-7/3}{-26/3} = \frac{-7}{-26} = \frac{7}{26}$$

$$\Rightarrow U_{33} = \frac{231}{26}$$

$$\text{So, } L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 7/26 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & -1 \\ 0 & -26/3 & 5/3 \\ 0 & 0 & 231/26 \end{bmatrix}$$

As

$$AX = b$$

$$LUX = b$$

$$(LU = A)$$

$$\text{Let } UX = Y \quad \text{--- (3)}$$

$$LY = b$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 7/26 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

$$y_1 = 3$$

$$2/3 y_1 + y_2 = -5 \Rightarrow y_2 = -7$$

$$1/3 y_1 + 7/26 y_2 + y_3 = 8 \Rightarrow y_3 = \frac{231}{26}$$

$$\text{from (3). } UX = Y$$

$$\begin{bmatrix} 3 & 1 & -1 \\ 0 & -26/3 & 5/3 \\ 0 & 0 & 231/26 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \\ 231/26 \end{bmatrix}$$

$$3x + y - z = 3 \Rightarrow x = 1$$

$$-26/3 y + 5/3 z = -7 \Rightarrow y = 1$$

$$\frac{231}{26} z = \frac{231}{26} \Rightarrow z = 1$$

Crout's Method

23/2/24

If we take $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$ $U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$

lower triangular $\rightarrow$ Unit upper triangular

egs $3x + y - z = 3$

$2x - 8y + z = -5$

$x - 2y + 9z = 8$

This system of linear eq. can

be written as

$AX = b$

Let $LU = A$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix}$$

By Comparing

$l_{11} = 3$

$l_{21} = 2$

$l_{31} = 1$

$u_{12} = 1/3$

$u_{13} = -1/3$

$l_{22} = -26/3$

$u_{23} = \frac{1 - (2)(-1/3)}{-26/3} = -5/26$

$l_{32} = -2 - (1)(1/3) = -7/3$

$l_{33} = 9 - (1)(-1/3) - (-7/3)(-5/26) \Rightarrow l_{33} = \frac{231}{26}$

So,

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -26/3 & 0 \\ 1 & -7/3 & \frac{231}{26} \end{bmatrix}, U = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1 & -5/26 \\ 0 & 0 & 1 \end{bmatrix}$$

Let $AX=b$ then $LUX=b$ (put $LU=A$)

Let $UX=Y$ (3) then $LY=B$

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & -26/3 & 0 \\ 1 & -7/3 & 2^{31}/26 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

Sol, $3y_1 = 3 \Rightarrow y_1 = 1$

$2y_1 - \frac{26}{3}y_2 = -5 \Rightarrow y_2 = \frac{-7 \times 3}{26} = \frac{+21}{26}$

$y_3 = 8 - y_1 + \frac{7}{3}y_2 \Rightarrow y_3 = 1$

From (3) $UX=Y$

$$\begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1 & -5/26 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ +21/26 \\ 1 \end{bmatrix}$$

Sol, $z = 1$

$y = 1$

$x = 1$

Best of Luck Y'all 🙌

Regards,

